Yifan Li

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Education

Princeton University Sep. 2024 – Present
Ph.D., Atmospheric and Oceanic Sciences

University of Reading & Nanjing University of Information Science and Technology Sep. 2020 – June 2024
Dual B.Sc., Atmospheric Sciences (**First Class Honours**)

Publications

Li, Y., & Cao, J. (2026). Diagnosing the increase in East Asian summer monsoon precipitation under a low-emissions scenario from an energy perspective (*Manuscript in Preparation*).

Using multiple climate model simulations together with moisture budget and moist static energy budget analyses, we diagnose the increase in East Asian summer precipitation under the SSP1-2.6 scenario relative to other scenarios. Our results suggest that reduced anthropogenic aerosols over East Asia increase surface solar radiation and evaporation, thereby enhancing convective activity and large-scale upward moisture transport, intensifying the East Asian summer monsoon circulation, and increasing rainfall.

Li, Y., Lin, M., Naik, V., & Horowitz, L. (2026). Recent tropospheric ozone trends in the northern mid-latitudes: roles of Asian emission reductions and climate variability (*Manuscript in Preparation*).

Using high-elevation observations and the GFDL climate model, we show that reductions in Asian emissions are the primary driver of the decreasing free-tropospheric ozone trend over the western US. Stratosphere-troposphere transport (STT) strongly modulates the interannual variability of free-tropospheric ozone over the western US. Climate variability modulates ozone anomalies over the western US by influencing STT through changes in the meandering of the polar jet, rather than the Pacific subtropical jet.

Li, Y., & Shen, X. (2026). Using causal inference to quantify the roles of tropospheric wave sources and stratospheric preconditioning in sudden stratospheric warmings (*Manuscript in Preparation*).

Apply a physically based causal inference framework to quantify the relative roles of tropospheric wave sources and stratospheric preconditioning in influencing the occurrence of sudden stratospheric warmings (SSWs). Our results indicate that both factors play essential roles in the occurrence of SSWs. Approximately one-third of the effect of tropospheric wave sources and one-half of the effect of stratospheric preconditioning on stratospheric polar vortex variability are mediated by stratospheric wave-activity fluxes.

Marais, E. A., Kelly, J. M., Vohra, K., **Li, Y.**, Lu, G., Hina, N., & Rowe, E. C. (2023). Impact of legislated and best available emission control measures on UK particulate matter pollution, premature mortality, and nitrogen-sensitive habitats. *GeoHealth, 7*(10), e2023GH000910. https://doi.org/10.1029/2023GH000910

Talks & Presentations

Asian and Stratospheric Influences on Springtime Background Ozone Over the Western US During 1980–2025
Intermountain West Ozone Exchange · Session 8 · Colorado, USA · October 27, 2026
Oral Presentation · Speaker · Upcoming
Received a **$2,200 student grant** from the Colorado Air Quality Enterprise.

Asian and Stratospheric Influences on Springtime Background Ozone Over the Western US During 1980–2025
GFDL–Princeton 71st Anniversary Symposium · Princeton, New Jersey, USA · October 23, 2026
Poster Presentation · Presenter · Upcoming

Estimating High-Resolution Top-Down Nitrogen Oxides Emissions for Improving Air Quality and Public Health in China
EGU General Assembly 2026 · Session AS4.9 · Vienna, Austria · May 5, 2026
Oral Presentation · Co-author
Conference abstract: *EGU26-1055*.

The Health Burden of Air Pollution in the UK: A Modelling Study Using Updated Exposure-Risk Associations
UK Public Health Science 2023 · London, UK · November 24, 2023
Poster Presentation · Co-author
Conference abstract published in *The Lancet, 402(Suppl. 1), S66* · Poster.

Selected Honours & Awards

Outstanding Graduation (only **5** out of 340 at Academy or **1** out of 43 at Department)	June 2024
First Class NUIST Scholarship (**top 5%**)	Oct. 2021 & Oct. 2022 & Oct. 2023 & June 2024
NUIST Merit Student Honor (**top 5%**)	Oct. 2021 & Oct. 2022 & Oct. 2023
Second prize for the 8th Internet+ National College Student Innovation and Entrepreneurship Competition (**National-level Award**)	June 2023
Third prize for National Undergraduate Research Opportunities Programme	May 2022
Second prize for NUIST 16th Mathematical Contest in Modelling	May 2022
Third prize for China Meteorological Modernization Science and Technology Innovation Competition (**National-level Award**)	May 2021
Third prize for Challenge Cup Academic Science and Technology Competition	Mar. 2021
Third prize for the NUIST 4th Advanced Mathematics Competition	Mar. 2021

Teaching Assistant Experience

Skills for Environmental Science module for Prof. Jing Yang	Mar. 2023 – Aug. 2023
Linear Algebra module for Prof. Yanmei Xue	Sep. 2022 – Mar. 2023

Service

Assistant Counsellor for Undergraduates, Class of 2029	Sep. 2021 – Nov. 2021
Epidemic Prevention Volunteer	Aug. 2021 – Oct. 2021